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Product Policy

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1 INTRODUCTION

How does a company decide the types of products or services it will sell? More specifically, how does it decide how many product lines it will develop, and how many different items to offer within each line? These fundamental decisions constitute a company's product policy. Over time, management must revisit these decisions and adapt the line through addition, deletion, or modification of products for sale.

Consider how the products offered by Bose Acoustics developed and evolved. In 1964, Amar Bose founded his company after he purchased stereo speakers for his own use that he found disappointing. Bose Acoustic's mission would be to create "better, more lifelike sound."[1] Four years later, its initial product offering, the 901 speakers, garnered critical acclaim. Bose later expanded its product offerings to include the less expensive, smaller 301 Bookcase speaker.

Then, in 1978, Amar Bose again had a disappointing audio experience, this time with the airline-provided headphones he received on a transatlantic flight. This launched the company into research on noise-canceling headphones, resulting first in products for the aviation and military markets, and later in the Quiet Comfort line of headphones for the consumer market. Over the years, the Quiet Comfort line was improved and expanded to include in-ear, around-ear, and on-ear models. Throughout, Bose retained its focus on "better, more lifelike sound" rather than on price: in 2014, for example, its noise-canceling line started at $299.95, about 30% higher than well-known brands such as Sony and Beats. Later, Bose moved beyond the audio domain to "leverag[e] the company's intellectual assets in broader ways,"[2] such as researching whole-body vibration and designing a seat for heavy-duty truckers.[3]

Note how each point in the Bose story was, in effect, a decision about what the company's product policy would be. First, Bose decided what business it wanted to be in; initially, this was home speakers for consumers. Second, it made a targeting decision—to serve both home dwellers with the 901 and apartment dwellers with the smaller 301 speakers. Over time, Bose expanded the kinds of markets it addressed—moving to professional sound and later beyond sound with its truck-seat design. Finally, for each item offered, such as the 301 speakers, Bose had to set the optimal product design given the market to be served, making trade-offs between cost and performance along the way.

The Bose product policy decisions made over decades have today resulted in its product mix—the set of all products the company offers. **Exhibit 1** shows three key policy decisions that companies like Bose make to arrive at such a product mix.

EXHIBIT 1 Product Policy Decisions

In the exhibit, we see a company's three key decisions:

1. **Product mix breadth** refers to the variety and number of **product lines** offered. To arrive at the product mix breadth, managers must ask themselves: How many columns are there going to be (how many product lines will be offered?)? What is the relationship, if any, between the lines?
2. **Product line depth** is the number of items in a given product line.[a] To arrive at the product line depth, managers will ask themselves: Within a given column, how many *rows* (items) will there be in the line? How is the line to serve customer segments of varying tastes or **willingness to pay**? (For an in-depth discussion of this topic, see *Core Reading: Segmentation and Targeting* [HBP No. 8219].)
3. **Product item design** refers to the product's design specifications. To determine the product item design, managers will ask themselves: What will the design or specifications be for each cell in the matrix (meaning each item within a product line)? Note that for a multi-item line, decisions about an item's design should be made in light of the design of other items to create an optimal overall offering to the market. An item can further be disaggregated into **stock-keeping units (SKUs)** as it is offered, for example, in a variety of package sizes.

In this reading we examine each of the three areas above to present the principles for guiding effective product policy decisions, with a particular focus

[a] Sometimes this is referred to as *product line length.* Either term is suitable, but we will use depth throughout.

on new products. The new product could result from a decision to introduce a product line or to increase the number of items in a line to more finely target the market.

We begin the Essential Reading by exploring product mix breadth, that is, the strategic decision about which businesses to compete in. We then examine how to set the product line depth correctly, that is, the line's length and spacing, or the degree of difference between items in the line. This is followed by a look at product item design, or the detailed level of optimally designing an item given its role in the product mix. Finally, we look at how to pull these tasks together—by managing the product line over time, commonly known as the product's life cycle.

2 ESSENTIAL READING

2.1 Product Mix Breadth

Firms differ greatly in the product mix, or variety, of products offered. For example, General Electric (GE) generated its $146 billion in sales for 2013 based on a product mix of 21 different product categories, including aviation (such as engines for Airbus, Boeing, and fighter planes), health care (PET and CT scanners), home improvement (GE silicone caulks), power and water (nuclear power plants), and transportation (locomotives). Coca-Cola Company, in contrast, generated its $46 billion in sales by concentrating on a single product category, beverages. The company is relatively consistent in its product form, from soft drinks and water to juice, energy drinks, and tea.

Why and when do companies decide to expand the number of product lines offered? Often the decision results from a specific favorable economic opportunity to draw on the company's existing skills, even if there is little linkage to the company's current business. Greater benefit, however, is usually derived from developing a new product that connects in some way with the company's existing products. Consider the following three ways that such connections might arise:

1. Undertaking a business whose profit stream will probably correlate negatively with the profit stream of existing businesses—thereby reducing the overall risk of the enterprise. For example, a beverage company developing a strong position in the water market helps offset risk of decline in the soda market if hydration habits move heavily to water.

2 Leveraging a key asset of the company that underlies the current product offerings. For example, Bridgestone's main product line is tires, and therefore the company has appreciable expertise in rubber technology. The company pursued a new application of this competency by designing rubber buffering systems for constructing buildings that are more earthquake-proof, particularly in Japan. In another example, Procter & Gamble (P&G) has a wide variety of products across four sectors that the company claims "share common technologies": beauty, hair, and personal care; baby, feminine, and family care; health and grooming; and fabric and home care. Because many of these products also share the same distribution system, P&G is able to leverage its established retailer relationships to market and sell its products.

3 Tapping into complementary-in-use products and thus enabling the firm to be a "total solution supplier." For example, Adobe acquired Omniture to develop a software line measuring the effect of marketing content to complement its leading position as provider of software for content development. Procter & Gamble explained the acquisition of the Gillette Company and its Oral-B toothbrush line by noting that the union of Oral-B and Crest placed P&G oral care as the market leader and the only major oral care company with a breadth of products across every category: toothpaste, toothbrushes, whitening, rinse, denture care, and floss.

In all these cases, managers must carefully analyze the interrelationship between product lines to ensure that positive links are taken advantage of and to guard against potential negative links—for example, competition for scarce resources that could damage overall product mix.

Once a company has determined its product mix breadth, it must then consider the second major decision it must make about its product policy: how many items it will include within each product line.

2.2 Product Line Depth

When Sealed Air Corporation began making protective packaging for items being shipped or mailed, it offered eight grades, or strengths, of bubble wrap. The lightest grade, suitable for protecting a lightweight, inexpensive item, was priced at $30 per 1,000 square feet, while the heavy-duty product, for heavier, more fragile, and expensive items, was priced at $141 per 1,000 square feet. Even with eight offerings of protective wrap, some customers requested products that more precisely met their needs.

If Sealed Air was to add a new item to the line, it had three options:

1. An upward stretch of the product line, adding an item at a higher price and performance level, able to meet the performance requirements of more demanding applications.
2. A downward stretch of the line, adding a lower-priced item still capable of satisfying the needs of some applications.
3. A product line fill, inserting an item of a price and performance level that fills a gap between two existing items.

In Sealed Air's case, the elements in the line were differentiated vertically—higher-priced items offered more protection. The reason there were eight items in the line was that, for some applications, less protection was perfectly adequate. Over time, Sealed Air perceived more opportunity at lower price and performance points, so it executed a downward stretch of the line.

In contrast to Sealed Air's vertically differentiated product line, items in a line can be horizontally differentiated at the same quality level to accommodate particular tastes. For example, Kellogg's offers 18 items in its Cheez-It crackers line, with flavors ranging from Original Cheddar, Swiss, Provolone, and Colby to Hot and Spicy. The cheez-it.com website even offers a tool to help customers find "the perfect Cheez-It snack" depending on their taste for (1) classic versus fun, (2) mild versus hot, and (3) at home versus on-the-go.

Product line depth decisions are driven by how many different segments within the target market the firm chooses to serve. Unless the product line is well managed, management's desire to serve customers (or retail partners) with precisely the right product for them can lead to excessively long product lines. Consider, for example, whether Samsung's sales would decline if it offered only 48 different television set models in the United States rather than its current 49. The answer is "not likely"—and thus we could say that perhaps Samsung need not have expanded the depth of its television product line to the extent that it has.

Product line depth decisions, therefore, must be carefully analyzed and planned—and should be guided by a vision for what the product line is to become over time. **Exhibit 2** presents a list of nine key considerations in the product line depth decision-making process. These elements affect product line architecture, whether for a vertical line like Sealed Air's or a horizontal one like Cheez-It's.

EXHIBIT 2 Product Line Depth Considerations

1. **Customer heterogeneity and requirements specification**
2. **Ability to configure the offering to the segment**
3. **Competitive impact**
   Preempt?
   Point of sustainable differentiation?
4. **Legitimization**
   Of product type?
   Of price point?
5. **Category size impact**
6. **Net impact on own margins**
   Cannibalization?
7. **Brand equity**
   Enhanced or diluted overall?
   Ability to support the extension
8. **Cost of variety vs. scale opportunity**
   R&D spend to develop
   Production/supply chain management impact
9. **Collaborator (e.g., distributor or retail) reaction**

Source: Adapted from "Extend Profits, Not Product Lines" by John A. Quelch and David Kenny, *Harvard Business Review*, September/October 1994. Copyright © 1994 by the Harvard Business Publishing Corporation; all rights reserved.

Quelch and Kenny presented a convincing warning about offering too many products in a line in their article "Extend Profits, Not Product Lines."[4] The following section summarizes and extends their arguments as we look at each of the nine considerations listed in **Exhibit 2** more closely.

1. Customer Heterogeneity and Requirements Specification

The first consideration in the product line depth decision is the heterogeneity or segmentation of the customer base and the "tightness" of the product specification required to meet customer needs. For example, Caterpillar makes heavy equipment such as bulldozers and tractors, which are typically used by sophisticated buyers with a particular job or set of jobs to do. Therefore, a product that almost meets the specification won't be good enough. This creates an incentive to extend the line. As a result, the Caterpillar product equipment line has more than 300 machines, including 19 bulldozers. With these 19 offerings, Caterpillar noted: "From large to small, mining to finish work, you are certain to find a Cat Dozer to match your needs."[5]

If consumer requirements are relatively homogeneous across the market and those requirements can be met by products of varied specifications, the line can be shorter. (For example, a basic individual-cup coffeemaker suits most consumers' needs, even though an individual consumer might value additional features more than another.

2. Ability to Configure the Offering to the Segment

Even though a firm may identify a customer segment it wishes to target with a particular product, its strategy for product line depth depends on its ability to configure and position the product so that the intended customers see the product as being made for them. What is the mapping of products to segments or applications? In some cases this is fairly clear: for example, diet soda and reduced-fat foods are aimed toward calorie-conscious consumers. In some situations, however, an inability to communicate this mapping can result in customer confusion. For example, when Acushnet first extended its Titleist golf ball line with the ProV1x, matching the price point of the ProV1 already on the market, it faced a challenge in communicating to its target market. This was not an issue for professional players, since a regular part of their job was extensive testing of equipment to determine which product suited them best, but many serious amateurs who were willing to pay a top price were confused. A competitor, Callaway, took advantage by advertising that it was simpler to buy just the one offering at the top of the Callaway line, rather than risk buying the wrong Titleist ProV1 offering.

A "good-better-best" vertical differentiation strategy, such as the one Sears uses for its car batteries, exemplifies a straightforward mapping of a product to a customer's willingness to pay. In contrast, consider GM's job of positioning its three Buick sedans. There was a clear price-point difference of about $5,000 between models:

Buick Verano: $23,700

Buick Regal: $29,690

Buick LaCrosse: $33,535

The problem, however, was that the benefits of the added consumer expense in trading up to the higher-priced item was unclear. GM marketing communications positioned the Verano as a modest car of "pleasant surprises," in which one could "feel completely at ease." Moving up, the Regal was where "exhilaration comes standard." What, then, to say about the LaCrosse? GM's choice of the tagline "Designed with One Thing in Mind: You"[6] illustrates the challenge in distinguishing a third segment in the $23,000 to $33,000 range. Was there room enough in that price range for three distinct models?

3. Competitive Impact

A company may try to achieve competitive advantage with its product line. For example, it may (1) *preempt* a competitor or (2) achieve a point of *sustainable differentiation* with a product line expansion. It can preempt a competitor by expanding its product line and serving a greater range of customer segments. Preemption occurs when a product serves a market segment in such a way that there is no room for a second company to profitably serve that segment. For example, the story is often told that Japanese companies were able to dominate the motorcycle market in the United States because of a missed preemption opportunity by competitors. When US company Harley-Davidson neglected the lower end of the market, Honda and Suzuki thus were able to enter the US market and establish a reputation for quality products and built a distribution system. This enabled them to later trade up to high-performance bikes (and, later, automobiles).[7]

A second incentive to expand a line can be the ability to serve a segment better than competitors can, using company skills to offer a differentiated product delivering superior value to a segment of the market. For example, Apple's iPhones are perceived by some as being simply better than competitors' offerings. Furthermore, broad product lines can make it difficult for smaller firms to compete by filling niches in the market, thereby achieving a point of sustained differentiation. For example, Sensodyne and Tom's of Maine were able to carve out niches in the toothpaste market by specializing, but the broad lines of Crest and Colgate limited incursion.

4. Legitimization

Extension of a product line by a leading firm into a *product type* or niche at a particular *price point* can legitimize that spot in the market, possibly to the benefit of a competitor or potential entrant. For example, consider the case of Sealed Air Corporation, discussed earlier. It was the leader in the flexible-wrap protective-packaging market; its "bubbles" were coated on the inside with a special material (saran, or polyvinylidene chloride) that promoted air retention and protective ability.[8] When a small competitor marketed an uncoated product at a lower price for less demanding applications, Sealed Air faced an important question: would expanding the line with an uncoated product legitimize that type of offering and the competitor's existence? Legitimization can also occur if a firm extends its line downward to a new price point. For example, Titleist's **brand equity** in the golf ball market was so strong that its offering the "DT Solo" at $28 per dozen (in contrast to $62 for its top-quality ball) signaled that a good-quality ball could be found for under $30 per dozen, to the potential benefit of competitors already operating there.

5. Category Size Impact

A key question in expanding the product line is whether a new item will increase demand for the product category overall. Quelch and Kenny suggest that while many line expansions seem to be justified on this basis, the facts seldom support the expansion idea. For example, they provide data showing that although the coffee and shampoo categories showed a 44% increase in number of items offered overall during a four-year period, both categories declined in overall dollar sales in real terms.

If a new item gains sales by means of category expansion, it is less likely to trigger a competitive reaction than if it has a "share takeaway from competitors" effect. Category expansion can stem from an increase in either the number of category users or the average consumption rate. Apple's hope for the 5c—a lower-cost addition to the iPhone line at the time of the release of the 5s—was not only to provide a competitor to Samsung and others but also to increase the sales of smartphones overall by means of the lower price.

6. Net Impact on the Company's Own Margins

Conceptually, a unit sale for a new item can come from one of the following three places: (1) an expansion of the category, (2) taking away a sale that would otherwise have been made by a competitor, or (3) taking away a sale that otherwise would have gone to an existing item in a company's line, that is, **cannibalization** of sales of a company's own items. Cannibalization is typically an issue in the case of a trade-down **product line extension** (an extension to a less expensive version of an existing product), since such products typically offer a lower margin than the existing elements of the line. Thus, any cannibalized units represent a decline in the sales margin that is generated.

Explore the effect of positioning on cannibalization and margins in **Interactive Illustration 1**. Altius is a golf ball manufacturer that plans to extend its product line. In the illustration, Altius already has two types of balls on the market. Balls are differentiated on two dimensions. The vertical dimension is overall quality, extending from economy to premium. The horizontal axis is distance, extending from short-range to long-range (referring to the distance a golfer could potentially hit the particular ball). Compared to those of competitors generally, Altius balls—shown at positions A1 and A2 in the illustration—are of relatively high quality and have a short range. Where should

Altius position its third ball? Playing the role of brand manager, find the product position between high and low quality, and short and long range. Find the best combination that balances market share gain, cannibalization, and profit margin. Note that golf balls marketed as premium enjoy higher margins (12%) than those marketed as economy (6%).

The illustration shows three different models. All three models assume that the new product takes away some market share from each existing product, based on how closely the new product was positioned to each. In addition, Model 1 assumes a propensity to take away greater market share from competitors than from other Altius balls. Model 2 demonstrates a brand loyalty effect, by assuming a greater propensity to take away the market share from other Altius balls than from competitors. Model 3 is a "fair share" model that assumes no such propensities; the market share taken by the new Altius ball from any existing ball does not depend on whose ball it is, but only on how close it is. How does your strategy change if your market conditions change?

INTERACTIVE ILLUSTRATION 1 Altius Golf Ball Positioning

Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2pIm2Lk

*Note: If the Interactive Illustration is not accessible, please see the PDF explanation of the Interactive **here**.*

In considering a line expansion, marketers must assess the source of volume for the new product to judge true financial impact and likelihood of competitive reaction.

7. Brand Equity

Two questions are critical here. First, in considering an expansion using an existing brand name, one has to judge the brand's ability to support that extension. For example, the Apple brand was clearly quite capable of supporting the extension of the iPhone 5 offering to include both the 5s and 5c. Conversely, when Toyota decided to go significantly upscale in its automobile line, it decided that the Toyota brand name could not support it and so created the new Lexus brand. Generally, this question is of more significance for trade-up extensions than trade-down or horizontal ones.

Second, one must address the effect of a **brand extension** on the equity of the overall brand itself—in other words, will it *enhance*, or build, brand equity (as was the case with American Express and its Gold and Platinum cards, given the quality of their delivery on those services), or will it *dilute* brand equity? Brand equity derives from the favorable brand associations that a consumer holds. Thus, even an extension of the brand to a trade-up situation can conceivably dilute brand equity. For example, one may perceive Bang and Olufsen speakers as a high-quality brand that, while expensive, represents good value—and is worth the price. Extension of the line to a new, higher-priced offering could upset these perceptions about value and worth—even if the consumer had never sampled the new speakers' sound. Similarly, brand extension of the horizontal type may detract from its brand equity if it is inconsistent with currently held associations. For example, Nike management admits that it made a mistake in its early days by extending the Nike brand to casual shoes. For details, see *Core Reading: Brands and Brand Equity* (HBP No. 8140).

8. Cost of Variety versus Scale Opportunity

In this factor, one must assess the overall effect of line extension on costs. On the plus side, if an extension increases unit sales of the line and the items in the line are derived from the same platform, the added sales can drive down costs across the line. On the other side, however, several potential problems can arise. First, incremental *research and development costs* can be significant, especially if the new item presents technological challenges. Second, there may be hidden costs in production, since more items means shorter production runs and more changeovers. In addition, inventory carrying costs may increase and the *supply chain* may become more difficult to run efficiently. Quelch and Kenny report that the "cost of variety"—that is, the added cost of producing the full line versus just the most popular item—is 25–45% in some consumer packaged-goods situations.

9. Collaborator Reaction

Finally, companies must assess how partners and collaborators, such as distributors and retailers, will react to how new items are marketed. On the one hand, these partners may see a new item positively, and as an investment in the brand to keep it growing. On the other hand, the addition of a new item, even if it boosts sales of the line overall, can lead to a decrease in sales velocity per SKU—a metric that collaborators may use in evaluating the brand and making stocking decisions.

Taken together, the nine considerations presented above argue for careful planning of product line evolution and length, rather than myopically adding a line on the basis of a belief that expanding consumers' options is always a good thing. By using the checklist we have just outlined, and regularly asking whether an item should be deleted from the line, marketers can feel more confident in their overall product architecture.

A line extension should be both desirable for some segment of customers, by virtue of its distinctiveness and the value it offers, and deliverable by the company—communicable so that the segment adopts it. This combination can help make it profitable for the company when all existing product interactions are considered.

2.3 Product Item Design

Once all the decisions have been made to determine the depth of a company's product line, management then moves on to the last major decision regarding its product policy: the actual specifications for the items it will develop and sell within each product line (or each cell within the overall product line matrix, as shown in Exhibit 1).

At this point it is useful to step back and view a product's position as one element in a company's overall marketing mix, as illustrated in **Exhibit 3**. (For more on the framework described in Exhibit 3, see *Core Reading: Framework for Marketing Strategy Formation* [HBP No. 8153].)

EXHIBIT 3 Schematic of the Marketing Strategy Formation Process

Source: Reprinted from Harvard Business School, "Framework for Marketing Strategy Formation," HBP No. 8153, by Robert J. Dolan. Copyright © 2014 by the President and Fellows of Harvard College; all rights reserved.

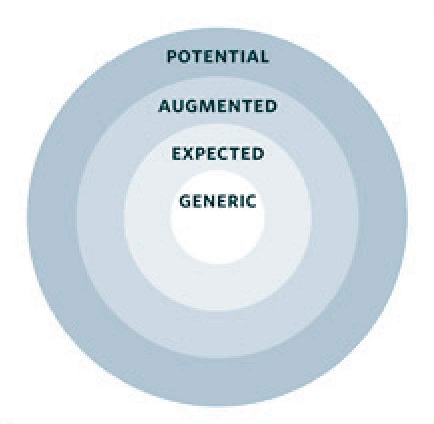
The exhibit depicts three interrelated elements in the marketing strategy formation process: analysis, decisions, and outcomes. The middle of the diagram depicts two major decisions an organization must make. First is the aspiration decision, or what the firm hopes to achieve in the market (that is, setting out the product positioning to be achieved in the mind of the target consumer, which offers strategic guidance to product development). For example, Volvo's choice of a safety positioning influences its car design. Second is the action plan decision, commonly known as the marketing mix of the four Ps—product, promotion, place, and price. As the exhibit illustrates, the product has a key role in creating customer value, but it must fit with two other value-creating elements, promotion and place (how/where the product is distributed).

The product plays a special role in that it is the means by which the actual customer's "problem is solved" and value of customer benefits delivered. Promotion plays a supporting role by helping make the customer aware of the product and its features; and place decisions are intended to make the product conveniently available.

When the full dimensions of a target customer's wants and buying behavior are understood, the best value-creation process typically involves more than just a tangible product or service. Starbucks, for example, augmented a good-tasting cup of coffee with ambiance, creating a relaxing "third place" (beyond home and work) for the consumer to take a break. In a business-to-business context, Northern Telecom achieved great success with a telephone system for small businesses by studying the business model and challenges of those who distributed and installed such systems. All expected a phone that was easy to install and reliable, but additional value could be achieved if the distributor's inventory risk could be mitigated. As a result, Northern Telecom introduced a new system that was not only a high-quality system, but also one that Northern Telecom would deliver to the installation site upon 24-hours notice. This meant that distributors had no inventory management task or risk.[9]

Theodore Levitt has described four different levels of products to be considered, as shown in **Exhibit 4**.[10] Let's look at each in turn.

EXHIBIT 4 Product Structure



Source: Adapted from "Marketing Success Through Differentiation—of Anything" by Theodore Levitt, *Harvard Business Review*, January–February 1980. Copyright © 1980 by Harvard Business Publishing Corporation; all rights reserved.

Generic product. The generic product is a combination of elements required to participate in the market, or the "table stakes" of the game. Some refer to this as features needed to deliver the core benefit of the product. For example, a pen has to disperse ink evenly across the page and a cola drink has to provide the expected sensation in the consumer's mouth.

Expected product. The expected product includes the generic product plus the minimum that the customer normally expects from a product in the category. For example, in the luxury car market, a minimum of a four-year/50,000-mile warranty is expected; in online banking, it is 24/7 support.

Augmented product. The augmented product goes beyond the expected product to include unexpected value-enhancing elements. To recognize opportunities here, it is useful to think about the customer's entire purchase-and-use experience. Levitt created this product terminology in the 1970s, and today smart marketers think about the overall customer experience. For example, a key part of Apple's value to customers is the high-level support, the Genius Bar, provided free as part of the ownership experience.

Potential product. Finally, the potential product is an extension to the augmented product that includes "everything that might be done to attract and hold customers," according to Levitt's definition. The potential product is one that is augmented, but in a way that goes beyond what most people might conceive of as possible. For example, in the pest control industry, the range of businesses that dealt with unwanted insects and other pests was well established. However, "Bugs" Burger challenged the typical business model by focusing on customers, such as restaurants and hotels, for whom *elimination*, not control, was key. Burger called his company "Bugs" Burger Bug Killers, he guaranteed pest elimination, and he refunded all fees ever paid if another insect was ever found after his work was done. The value delivered to his chosen customers through this exercise of envisioning the potential product allowed him to charge the high prices necessary to support his operations.

Next we examine the particular characteristics that lead to winning products, followed by a look at the process through which new products are typically developed.

2.3.1 Characteristics of Winning Products

The key success factors of new products have been defined. The research programs of Robert Cooper[11] and Everett Rogers[12] have identified key characteristics marking successful new products. Cooper found that:

> a unique superior product—a differentiated product that delivers unique benefits and a compelling value proposition to the customer or user—is the number one driver of new-product profitability.[13]

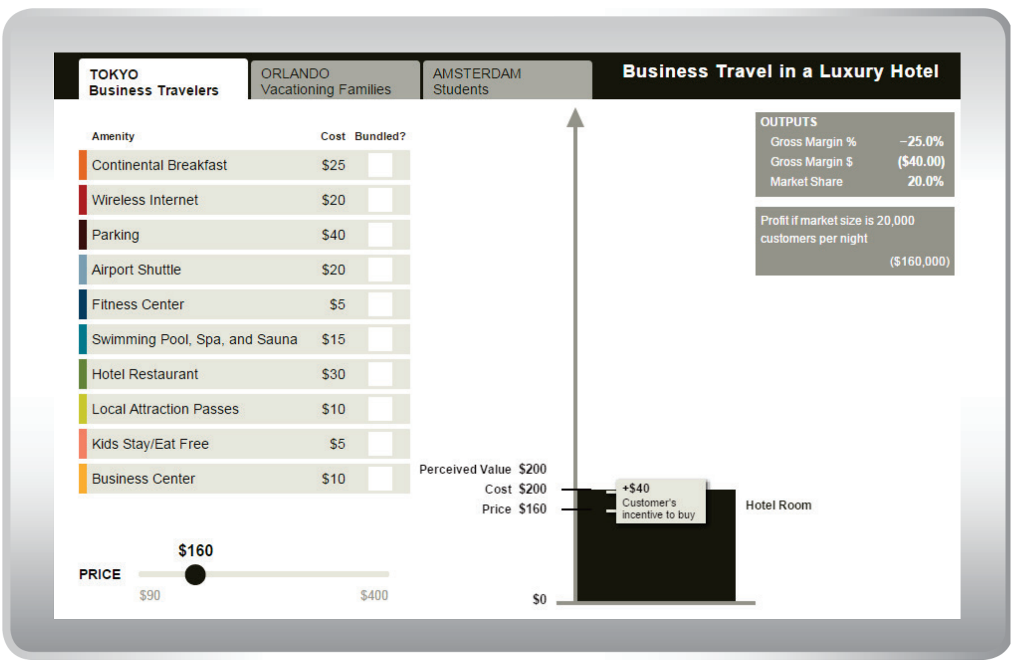
His empirical finding is that such products have success rates five times greater than those of other products. Expanding on this key principle, Cooper set out what unique and superior products tend to have in common. These products

1. are superior to competitors' products in terms of meeting users' needs
2. solve a problem the customer has with a competitive product
3. feature good value for the money and excellent price and performance characteristics
4. provide excellent product quality, according to customers' way of defining quality
5. offer features easily perceived as useful by the customer
6. offer benefits that are highly visible to the customer[14]

Note that many products and services—such as hotels and health clubs, to name just two—can be considered a bundle of characteristics or attributes. The attributes that are proven value drivers can be harder to identify in a market of mixed segments. Consider a hotel that has locations in Tokyo, Japan; Orlando, Florida; and Amsterdam, Netherlands, as shown in **Interactive Illustration 2**. This hotel is targeting a different market segment in each city—vacationing families, business travelers, and vacationing college students, respectively—and creates a bundle and price structure tailored to each. Constrained by amenity cost and customer perception, what are the highest value drivers for students exploring a foreign city? Vacationing families focused on child entertainment? Busy professionals? Which of the 10 possible value-adding amenities should each hotel offer, and what price should they charge for that bundle? Note: The model estimates the achieved market share for a given price/bundle combination by comparing price and perceived value.

INTERACTIVE ILLUSTRATION 2 Product Bundling

Scan this QR code, click the image, or use this link to access the interactive illustration: bit.ly/hbsp2TJ6dmU



Note: If the Interactive Illustration is not accessible, please see the PDF explanation of the Interactive here.

Another key aspect of a successful new product or service is how quickly and easily customers adopt it. Everett Rogers identified five attributes (see **Exhibit 5**) that explain the rate of adoption of an innovation by target consumers.

EXHIBIT 5 Rogers's Five Key Innovation Attributes

1 RELATIVE ADVANTAGE
What is the degree of improvement in the innovation relative to existing products?

2 COMPATIBILITY
How easy is it to integrate with current investments and ways of doing things?

3 COMPLEXITY
How easy is the innovation to understand? How easy is it to use properly?*

4 TRIALABILITY
How easy is it to try the product on a limited basis to assess its relative performance?

5 OBSERVABILITY
To what extent would use by an adopter be visible to others and hence create awareness of and knowledge about the innovation?

* The other four factors correlate positively with rate of adoption; this one correlates negatively.

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Importantly, Harvard Business School professor John Gourville has expanded on the work of Cooper and Rogers by identifying key psychological factors that create a reluctance to adopt an innovation among what he terms "stony buyers," who are reluctant to move from their current product choices.[15] Gourville also built on the Nobel Prize–winning work of Kahneman and Tversky to show that new-product development work must recognize the biases that consumers bring to adoption of new products, in particular consumers' reluctance, even beyond an economically reasonable level, to part with the product they currently use.

Gourville argues that most new products do not dominate the existing products on all attributes of importance. Instead, while a new product has to have gains in some attributes to be considered at all by consumers, in most cases, there is a loss when a new product is compared to existing alternatives. For example, online grocery shopping provides the convenience of home delivery, but at a cost of not being able to see, feel, and select the freshest produce. The key point here is that, as the Kahneman and Tversky research demonstrated, "losses have a far greater impact on people than similarly sized gains."[16] Gourville's advice to managers is to accept the unfortunate truth that "it's not enough for a new product to be simply better. Unless the gains far outweigh the losses, consumers will not adopt."[17] One proven way to increase the odds that a new product will succeed is to create an entirely new category. Keurig, for example, redefined the coffee category with its single-serving coffee pods, and Anheuser-Busch InBev redefined the spirits category by launching a margarita and malt-beverage hybrid (see **Video 1**).

VIDEO 1 Cashing in on Category Creation

Scan this QR code, click the icon, or use this link to access the video: bit.ly/hbsp2I7qfPo

The most direct and obvious implication of Gourville's work is that managers need to ensure that a new product's relative advantage over existing alternatives is perceived as being significant, so it can withstand the weight of the losses. Most often this requires pursuing a strategy to minimize adoption resistance. Gourville suggests the following:

1. Make behaviorally compatible products—for example, Toyota's Prius engine allows the vehicle to be operated much like a gas-only car.
2. Target customers not currently wedded to an incumbent product—for example, Burton aimed its snowboards not at expert skiers who had spent years developing their expertise on skis, but at novices who were perhaps less devoted to skis and therefore more likely to adopt a new snow sport.

2.3.2 New-Product Development Process

Given that the hallmarks of successful new products are pretty well understood, a key question becomes: How does a company develop a superior, highly valued product in the first place?

The first part of answering this question is to recognize that there is no one answer; that is, the process of new-product development must be customized to each situation. It would be foolish, for example, for GE to apply the same process to designing its latest microwave oven as it would to a low-cost CT scanner for emerging economies.

Exhibit 6 shows the full spectrum and some characteristics of product-development opportunities in different companies, including improvements on existing products to radical innovations.

EXHIBIT 6 New Product Spectrum

TYPE: Incremental improvements of existing product

Examples

Kitchen Aid Food Processor with Dicing Attachment
Samsung Refrigerator with Automatic Sparkling Water Dispenser
Bosch Dishwasher with Speed Perfect Mode (cutting run time by 30 minutes)

TYPE: Expansion of existing product line

Examples

Diet Coke (1982), Coke Zero (2004), Caffeine Free Coke Zero (2013)
Tanqueray 10 Gin (in more expensive direction)
Marc by Marc Jacobs (in less expensive direction)

TYPE: New product to firm, but not to world

Examples

Zantac (after Tagamet for ulcers)
Cialis (after Viagra for erectile dysfunction)

TYPE: New-to-world product/radical innovation

Examples

Tagamet
Viagra
GE CT Scan
Segway Human Transporter
Corning Optical Fiber

At the top of the spectrum are incremental improvements of existing products. Even when small (for example, a new flavor or package size), the change must be appropriately managed. While not offering the excitement of the breakthroughs at the bottom of the spectrum, these improvements, aggregated over the years, can be critical to a firm's competitive position and cost levels. Product development efforts here are typically guided by the firm's knowledge of the market from having been a participant in it. This intimacy with the market indicates valuable directions of research, and technological uncertainty is low. In contrast, at the bottom of the spectrum, radical innovations are generally characterized by high degrees of uncertainty. A company may not even know the market that will ultimately prove most receptive to a new product; therefore, it might not even know whom to survey as potential customers, if such a survey was needed for the development process.

While the different conditions surrounding each type of new product necessitate a different approach for each, there are important commonalities when it comes to developing any new product, such as the following:

1. The voice of the customer is heard throughout the process—albeit by different methods for different circumstances.

2 There is a process of "spiral development," in which versions of a product are built and tested in the lab and with customers. Feedback is then recorded and used to revise the product, the specifications of the target market, and the go-to-market plan.

For products toward the top of the spectrum, classic market research techniques such as focus groups, surveys, concept tests, and product-use tests are widely used. Consumers and firms share a vocabulary, and consumers can react to product ideas with useful feedback.

Another particularly useful research procedure in developing products near the top of the spectrum is the concept test, in which participants are given a description of the product and asked their reaction. For example, Colgate conducted a concept test (specifically a ***BASES test*** conducted by Nielsen) on its Max Fresh toothpaste with 213 US respondents. See the concept description in **Exhibit 7**.[18]

EXHIBIT 7 US "Colgate Max Fresh" Concept Statement

INTRODUCING

Colgate Max Fresh Toothpaste

Infused with Mini Breath Strips

Experience a whole new dimension of freshness

When you brush your teeth, you expect to freshen your breath—now it's time to raise your expectations so you can leave the house with that ready-for-anything confidence.

When you brush with Colgate Max Fresh whitening toothpaste infused with mini breath strips, you don't just get fresh breath, you experience a whole new dimension of freshness.

The mini breath strips are packed with mouthwash freshness. As you brush, they instantly dissolve, releasing an extra rush of breath freshening power so you can be confident that you're ready to take on the day.

New from Colgate, the only whitening toothpaste infused with mini breath strips. Experience a whole new dimension of freshness.

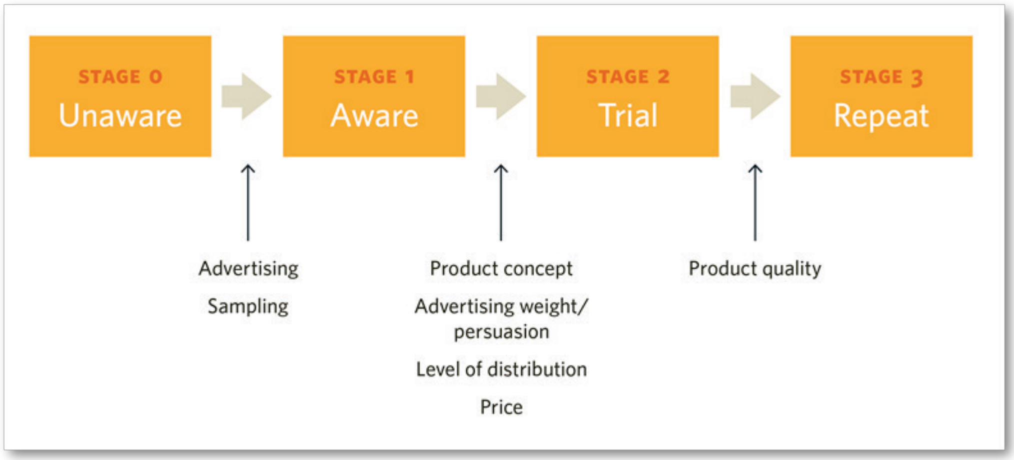
Source: Reprinted from Harvard Business School, "Colgate Max Fresh: Global Brand Roll-Out," HBS No. 508-009, by John A. Quelch and Jacquie Labatt-Randle.

In the test, marketers collected from participants the following key measures:

1 Purchase intent, ranging from "definitely would buy" to "definitely would not"
2 Perceived likeability
3 Perceived value
4 Perceived uniqueness

BASES (and other models such as **ASSESSOR**) use a simple **hierarchy-of-effects model**, as shown in **Exhibit 8**, to forecast the sales volume for a product before market introduction.

EXHIBIT 8 Hierarchy-of-Effects Model[b]



The model posits that consumers move through successive stages by the various market elements shown. This model breaks down the larger problem of forecasting sales to three smaller problems: predicting the percentage of customers who will become aware of the product; predicting the percentage of those who are aware of the product and who will try it; and predicting the percentage of those who try the product and will become repeat customers. Repeated use of such models, and the opportunity to compare actual and predicted share levels, has established an impressive record for this approach.

The general thinking is that the more one moves to the bottom of the spectrum (from incremental to radical new products) and the uncertainties portrayed there, the simple approach of "talk to representative customers and be directed by what they say" loses its power. For example, Lynne, Morone, and Paulson studied "discontinuous innovations" at Corning, General Electric, and Motorola. Their finding was that when the companies used conventional methods such as customer surveys and concept tests, their significance was limited: "In no instance was [the conventional market research] the critical factor in the decision making." They did find that the companies sought to hear the voice of the customer but in a fundamentally different way. The researchers called the adopted approach the "probe-and-learn" process, the process consists

[b] This model was inspired by Robert J. Lavidge and Gary A. Steiner's classic article, "A Model for Predictive Measurement of Advertising Effectiveness," *Journal of Marketing* 25 (October 1961): 59–62.

of "probing potential markets with early versions of the products, learning from the probes, and probing again. In effect, they ran a series of market experiments—introducing prototypes into a variety of market segments." This approach, the researchers reported, was experimental, as opposed to the more analytical approach of conventional methods. The probing was an experiment that provided a learning opportunity on both the market side (which segments were receptive to which features) and the product side (how the technology functioned and whether it could be scaled up).

Finally, the researchers stressed that probing and learning is an interactive process. This is similar to Robert Cooper's statement about a second general hallmark of the effective process of spiral development. Accordingly, Cooper developed the Stage-Gate model "to provide a conceptual and operational road map that moves a new-product project from idea to launch."[19] In this model, a product idea goes through the sequential stages seen in **Exhibit 9**.

EXHIBIT 9 The Stage-Gate Model

Source: Product Development Institute, Inc., "Stage-Gate®—Your Roadmap for New Product Development," http://www.prod-dev.com/stage-gate.php. Reproduced by permission.

However, Cooper warns against the process becoming "too rigid and linear." Rather, he suggests the following guidelines to this spiral approach:

- Build something (even if it's only a model)
- Get it in front of customers and obtain their reactions—likes, dislikes, and purchase intent
- Get feedback on what needs to be changed
- Revise—update and set up the next iteration of "build-test-feedback-and-revise"

This approach overcomes the fact that "customers don't really know what they are looking for until they see it or experience it."[20]

Video 2 describes a similar approach, called the "Feedback Loop," which offers innovators a way to manage the uncertainty that comes with any product development. By quickly cycling through a loop of (1) building an experimental version of the product, (2) testing it, and (3) gathering feedback from users on how to improve it, companies avoid becoming mired in endlessly perfecting an untested innovation.

VIDEO 2 The Feedback Loop

Scan this QR code, click the icon, or use this link to access the video: bit.ly/hbsp2ulgnjb

2.4 Managing the Product's Life Cycle

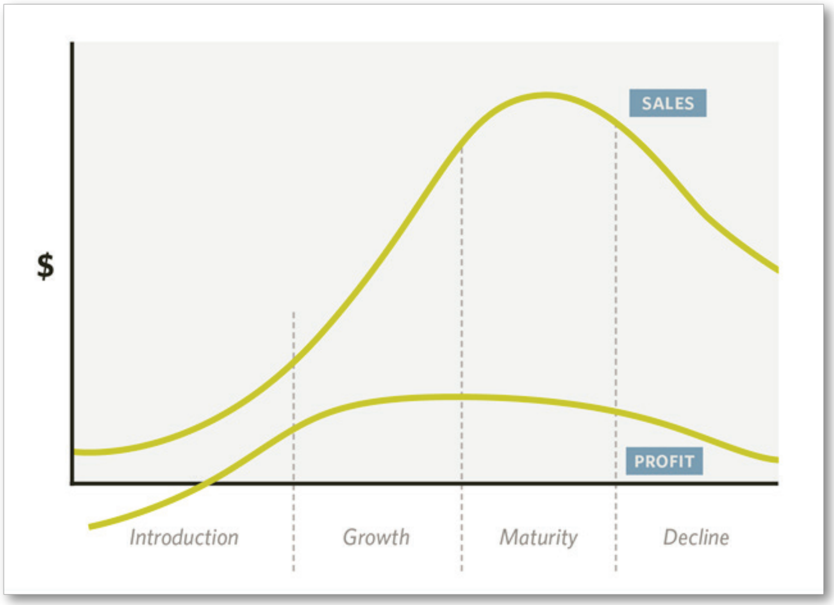
Once a new product is launched, attention turns to managing the product's life cycle. Products vary greatly in their longevity. Some flop quickly and are withdrawn from the market. Others last hundreds of years, and product strategy is adapted to market conditions. For example, the wine of Château Margaux dates back to the sixteenth century. In 1855 it was designated one of the four "premier crus" in Bordeaux, signifying top quality.[21] A bottle of Grand Vin du Château Margaux from a good year sold for thousands of dollars. In 1906 the château allocated some grapes to the making of a second, lower-quality level, "Pavillon Rouge." In 2013, to preserve the quality of the top wine, the châteaux designated even a smaller percentage of its grapes to it, instead introducing a third-quality wine from the châteaux's vineyards.

While some products never make it and others seemingly last forever, most follow a pattern marked by four stages:

1. Introduction
2. Growth
3. Maturity
4. Decline

The typical sales-and-profit path through these stages is shown in **Exhibit 10**.

EXHIBIT 10 Typical Sales and Profit Path



Consider Zantac, the acid-blocking ulcer treatment introduced by Glaxo in 1983.[22] Glaxo was second to market with an acid blocker, Smith Kline having pioneered the category with Tagamet six years earlier. Zantec offered more convenient dosing and fewer side effects than Tagamet, but early on Glaxo faced a significant marketing challenge in communicating this to prescribers in a compelling enough way to induce a switch in prescribing behavior. A massive investment in a "detailing" effort (a sales force backed by scientific support visited doctors' offices) was made to get through the awareness and trial stages. Even with this support, it was not until late 1987—four years after launch—that Zantac's sales rate matched that of Tagamet.

The growth phase continued through the mid-1990s as Zantac became the best-selling drug in the world, with sales reaching $2 billion in the United States and $4 billion worldwide. With the drug well-established in the market, promotional spending was not as crucial. Moreover, as the product's value became documented through patient use, Glaxo increased its price by 34% in real terms over its first decade on the market.

The decline stage for Zantac began as two additional acid blockers came on the market, and a new ulcer-treatment class of drugs, called proton pump inhibitors, was discovered and commercialized. By 1997 one of these new drugs, Prilosec, replaced Zantac as the world's best-selling drug. Furthermore, the patents underlying Zantac expired in 1997, and by 2000 sales had dropped to $200 million. By 2008, 25 years after its introduction, Zantac's sales were about $50 million, less than 1% of the parent company's sales.

Generally, the pattern that develops during each stage can be summarized as follows:

Introduction. Marketing expenditures are high as awareness and knowledge of product features must be established in the target market. Trial may be hard to induce if customers are satisfied with current options. Distribution must be built up.

Growth. With awareness and trial achieved, the product's performance is the dominant driver of sales. Success may induce the entry of new competitors and require promotional spending to defend against those competitors. Pricing is case specific. As happened with Zantac, price may be increased as the perceived value increases. In other cases, price declines (helped by cost declines) are enacted to reach a broader market.

Maturity. All feasible market segments have been reached and distribution channels built up. For nondurables, sales come from replenishment only, as household penetration has leveled off. For durables, population and income growth can be key drivers.

Decline. The rate of decline may still be a function of marketing efforts. A focus on hard-core loyals, who will make an effort to buy even at high prices and limited distribution, may be a profit opportunity. The discipline to phase out weak products is critical, however, lest the company and distribution channels become overwhelmed.

Products are at the heart of any marketing plan, since they are the mechanism for delivery of value to customers. Product decisions take many forms—ranging from strategic ones such as, "Should we be in the household appliance business?" (as GE asked in 2014—and answered "no"—after over 100 years in the business), to "How many cubic feet should our smallest microwave be?"

Listening to the voice of the customer is critical to all these decisions. The best method for obtaining customer feedback depends on the specific circumstances related to the product. Rapid prototyping—or getting a version of the product in front of a customer—is essential for making decisions about innovative products in particular. Then, once the product has launched, managers need to direct their attention to each stage of the product life cycle as customers and competitors evolve.

3 KEY TERM

ASSESSOR The ASSESSOR model is a pre–test market model that forecasts sales or market share (or both) for a new brand, provides a structure for evaluating alternative marketing strategies, and generates diagnostics that aid in improving the product.

BASES test A concept test that assesses a sample of consumers' reactions to a concept; a key measure is their expressed purchase intention. With this measure and others, such as perceived uniqueness of the product, BASES makes a market share forecast.

brand equity The value premium that a company realizes from a product with a recognizable name when compared to its generic equivalent.

brand extension Product line extension marketed under the same general brand as a previous item or items. To distinguish the brand extension from the other item(s) under the primary brand, one can either add a secondary brand identification or add a generic. A brand extension is usually aimed at another segment of the general market for the overall brand.

cannibalization A reduction in sales volume, sales revenue, or market share of one product as a result of the introduction of a new product by the same producer.

hierarchy-of-effects model 1. A concept related to the manner in which advertising supposedly works; it is based on the premise that advertising moves individuals systematically through a series of psychological stages such as awareness, interest, desire, conviction, and action. 2. An early model that depicted consumer purchasing as a series of stages including awareness, knowledge, liking, preference, conviction, and purchase.

product item design The detailed specification of a manufactured item's parts and their relationship to the whole. A product design needs to take into account how the item will perform its intended functionality in an efficient, safe, and reliable manner. The product also needs to be capable of being made economically and be attractive to targeted consumers.

product line A group of products or services marketed and positioned by an organization to one general market. The products have some characteristics, customers, or uses in common, and they may also share technologies, distribution channels, prices, services, and so on.

product line depth The number of product varieties (subcategories) within a product or service line or category. Companies with deep product lines focus on a specialized product mix or an expertise in a specialized service offering.

product line extension A new product marketed by an organization that already has at least one other product being sold in that product or market area.

Line extensions are usually new flavors, sizes, models, applications, strengths, and the like. Sometimes the distinction is made between near line extensions (very little difference) and distant line extensions (almost completely new entries).

product mix breadth The number of the product or service lines (categories) that a company offers to its customers.

stock-keeping unit (SKU) A specific unit of inventory that is carried as a separate, identifiable unit. In retailing it is the lowest level of identification of merchandise.

willingness to pay The maximum amount a consumer thinks a product or service is worth. This amount is considered when developing an asking price for products and services.

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